

Bosphorus Patent Technology Taxonomy: A Dual-Perspective Framework for Application- Oriented Patent Analytics

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Abstract—Traditional patent classification systems such as the International Patent Classification and the Cooperative Patent Classification provide standardized mechanisms for organizing patent documents according to technical fields. However, these systems may offer limited analytical resolution for rapidly evolving technologies, interdisciplinary inventions, and market-oriented use cases. This paper introduces the Bosphorus Patent Technology Taxonomy v0.1, an experimental dual-perspective framework designed for application-oriented patent analytics. The taxonomy classifies patents along two complementary dimensions: core technology and application domain. The current version consists of 23 main categories, including 10 core technology categories and 13 application-domain categories. Each category is supported by subcategories and bilingual keyword sets. In addition, the taxonomy is designed to support automated patent classification through embedding-based natural language processing. Using the `intfloat/multilingual-e5-large-instruct` model, taxonomy categories and patent texts can be represented in a shared semantic vector space, enabling zero-shot or few-shot classification based on semantic similarity. The framework is intended to complement, rather than replace, existing patent classification systems by providing a more flexible and market-aware analytical layer for technology intelligence, R&D strategy, competitive analysis, and investment evaluation. The taxonomy data, bilingual keyword sets, and automated classification scripts are open-source and available at: <https://github.com/vyzplay/bosphorus-patent-taxonomy>

Keywords—*patent analytics, technology taxonomy, patent classification, embeddings, technology intelligence*

I. INTRODUCTION AND MOTIVATION

Patent documents are not only legal instruments for protecting inventions; they are also detailed technical records of new products, processes, materials, software architectures, biological methods, electronic systems, and industrial applications. As a result, patent data are widely used in technology monitoring, competitive intelligence, R&D strategy, university-industry collaboration, investment analysis, and patent landscaping.

The analytical value of patent data depends strongly on how patent documents are classified and interpreted. The IPC, administered by WIPO, provides a hierarchical system of language-independent symbols for classifying patents and utility models according to areas of technology. WIPO states that a new IPC version enters into force each year on January 1, making the IPC a central reference for international patent organization and search [1].

The CPC, jointly developed by the EPO and USPTO, provides a more granular classification system based on the IPC. EPO describes the CPC as comprising nine sections, including A–H and Y, and approximately 250,000

classification entries. The Y section is particularly relevant for tagging emerging technologies and cross-sectional technologies spanning several IPC sections [2].

Despite their importance, IPC and CPC are primarily technical classification systems. They are highly effective for patent search, examination, and institutional data organization, but application-oriented analytics often requires additional interpretation. A single core technology may appear across multiple markets: for example, computer vision may be used in medical imaging, autonomous vehicles, agricultural monitoring, industrial inspection, or consumer electronics. Conversely, a single application domain, such as digital health, may combine artificial intelligence, sensors, biotechnology, cloud infrastructure, and embedded systems.

Recent patent analytics literature has increasingly emphasized the role of NLP, deep learning, and semantic representations in addressing such interpretive challenges. Surveys of patent NLP show that machine learning methods are now used for classification, retrieval, valuation, prior-art search, patent generation, and multimodal patent analysis [3], [4].

The Bosphorus Patent Technology Taxonomy is proposed as a complementary analytical layer. Its purpose is not to replace IPC, CPC, or WIPO technology-field mappings, but to help analysts read patent portfolios through two linked questions: What is the core technology? and Where is it applied?

II. RELATIONSHIP TO IPC, CPC, AND WIPO TECHNOLOGY FIELDS

The Bosphorus framework is conceptually aligned with existing patent classification infrastructures but differs in analytical purpose. IPC and CPC classify patent documents according to technical subject matter. WIPO’s IPC–Technology Concordance Table links IPC symbols with 35 fields of technology and is updated regularly to reflect IPC revisions [5].

The WIPO concordance is especially useful for macro-level patent statistics and country or sector comparisons. It answers the question: Which broad technology field does a patent belong to? The Bosphorus taxonomy asks a related but more application-oriented question: Which core technology does the invention rely on, and in which application domain does it create value?

This distinction is important because technological capability and economic use are not identical. WIPO’s own methodological discussion recognizes that technology classifications are a first step toward understanding economic activity, while a further connection between technologies and

sectors is needed to understand how technological competence translates into economic performance [6].

The Bosphorus framework therefore treats IPC, CPC, and WIPO technology fields as complementary signals. IPC and CPC codes may support validation or pre-filtering. WIPO’s 35 technology fields may support high-level benchmarking. The Bosphorus taxonomy adds a technology-market matrix that is better suited for portfolio interpretation, competitive intelligence, and application-oriented technology mapping.

III. THE BOSPHORUS TAXONOMY v0.1

The Bosphorus Patent Technology Taxonomy v0.1 classifies patents through two perspectives.

The first perspective, Core Technology, identifies the technical foundation of the invention. It asks whether the invention is primarily based on software, artificial intelligence, electronics, semiconductors, communication networks, photonics, robotics, materials, biotechnology, energy systems, or quantum technologies.

The second perspective, Application Domain, identifies the sector, market, or use context in which the technology is deployed. It asks whether the invention is applied in healthcare, mobility, industrial production, energy infrastructure, digital services, telecommunications, consumer electronics, construction, agriculture, aerospace, defense, chemicals, textiles, or everyday consumer products.

This dual structure allows a patent to be represented as a combination of technical and market-oriented labels. For example, a driver-fatigue detection system may be classified under artificial intelligence, computer vision, sensors, and embedded systems as core technologies, while being assigned to automotive, mobility, and transportation as its application domain. Similarly, a biosensor patent may combine biotechnology, microfluidics, electronics, and data processing, while being applied in healthcare, food safety, or environmental monitoring.

TABLE I. PERSPECTIVE A: CORE TECHNOLOGY CLASSES

Perspective A: Core Technology Classes
1. Software, Data, and Artificial Intelligence
2. Electrical, Electronic, and Embedded Systems
3. Semiconductors and Microelectronics
4. Networks, Transmission, and Connectivity
5. Optics, Photonics, and Imaging
6. Mechanical, Mechatronic, and Robotic Systems
7. Chemistry, Materials, and Surface Technologies
8. Biotechnology and Life Science Technologies
9. Energy and Environmental Technologies
10. Quantum Technologies

The resulting classification can be used to build technology-market matrices. A company’s patent portfolio may, for instance, be concentrated in artificial intelligence, sensors, and embedded electronics at the core-technology level, while being distributed across healthcare, mobility, and industrial manufacturing at the application level. Such

structures are not always directly visible from IPC or CPC codes alone.

TABLE II. PERSPECTIVE B: APPLICATION DOMAIN CLASSES

Perspective B: Application Domain Classes
1. Healthcare, Medicine, and Life Sciences
2. Automotive, Mobility, and Transportation
3. Industrial Production and Manufacturing
4. Energy, Infrastructure, and Utilities
5. Information Technology and Digital Services
6. Telecommunications Services and Network Operations
7. Electronics and Smart Consumer Devices
8. Construction, Buildings, and Built Environment
9. Agriculture, Food, and Bioresource Systems
10. Aerospace, Space, and Defense
11. Chemicals, Materials, and Process Industries
12. Textiles, Apparel, and Footwear
13. Living Spaces and Consumer Products

IV. AUTOMATED CLASSIFICATION ARCHITECTURE

Bosphorus v0.1 is designed not only as a manual taxonomy but also as a basis for automated classification. The proposed implementation uses multilingual text embeddings to represent both patent documents and taxonomy categories in a shared semantic space.

The preferred model for the initial implementation is intfloat/multilingual-e5-large-instruct. The E5 family was developed for producing single-vector text representations suitable for retrieval, clustering, and classification tasks, using weakly supervised contrastive pre-training [7]. The multilingual E5 models extend this approach to multilingual text pairs and include instruction-tuned variants, making them relevant for patent datasets that contain documents in multiple languages [7].

The proposed classification process consists of four steps.

First, Bosphorus categories, subcategories, and bilingual keyword sets are converted into vector representations. Each category is therefore treated not merely as a label, but as a semantic prototype.

Second, the patent title, abstract, independent claims, or selected description sections are embedded using the same model. This enables classification beyond exact keyword matching.

Third, cosine similarity is calculated between the patent vector and the category vectors. Core technology and application-domain categories are evaluated separately.

Fourth, classification decisions are made in zero-shot or few-shot settings. In zero-shot use, the system relies on category descriptions and keywords without requiring a large manually labelled dataset. In few-shot use, a small number of representative patents can be added to each category to strengthen category prototypes.

This design is consistent with recent embedding-based patent research. PatentSBERTa, for example, demonstrates

that patent-claim embeddings can be used to estimate patent-to-patent technological similarity and support semantic search and automated patent classification [8]. Scalable patent classification studies also emphasize the multi-label nature of patent classification, the difficulty of generalizing across many labels, and the value of using multiple textual views of patents [9].

IPC and CPC codes can also be incorporated into the architecture. For example, a patent's IPC codes may be mapped to WIPO's 35 technology fields and used as a validation signal. CPC subclass or subgroup information may support technical pre-filtering. Embedding similarity scores and institutional classification codes can therefore be combined in a hybrid classification pipeline.

V. LIMITATIONS AND FUTURE WORK

Bosphorus v0.1 is an experimental framework. It should not be interpreted as a finalized standard or a replacement for official classification systems. Several limitations remain.

First, category boundaries are inherently porous. Energy storage may involve energy technology, materials science, automotive applications, and infrastructure. Biosensors may combine biotechnology, electronics, microfluidics, healthcare, agriculture, and environmental monitoring. Multi-label classification is therefore essential.

Second, similarity thresholds are not yet calibrated. Cosine similarity scores indicate semantic proximity, but the threshold required for a reliable category assignment must be determined empirically. Thresholds may differ by category, patent section, and technology domain.

Third, zero-shot and few-shot classification are useful for early deployment but insufficient for high-stakes institutional use. A gold-standard dataset labelled by domain experts will be required to measure precision, recall, F1 score, and category-level reliability.

Fourth, the taxonomy must remain dynamic. Technology terminology changes quickly. New materials, algorithms, device types, standards, and regulatory concepts will require periodic updates to category descriptions and keyword sets.

Future versions should therefore focus on: expert-labelled validation data; category-level precision, recall, and F1 measurement; empirical threshold optimization; comparison with IPC, CPC, and WIPO technology fields; hybrid classification using both embeddings and official codes; and semi-automated keyword expansion using new patent publications and scientific literature.

VI. CONCLUSION

The Bosphorus Patent Technology Taxonomy v0.1 proposes a dual-perspective framework for application-oriented patent analytics. By separating core technology from application domain, it provides a structured method for interpreting patent documents as both technical artefacts and market-relevant innovation signals.

The framework is designed to complement established classification systems. IPC and CPC provide standardized technical classification, while WIPO's technology concordance table maps IPC symbols to 35 broader technology fields. Bosphorus adds an additional interpretive layer that supports technology-market mapping, portfolio

analysis, competitive intelligence, R&D strategy, and investment-oriented patent analytics.

The proposed embedding-based implementation makes the taxonomy scalable for large patent datasets. By combining multilingual semantic embeddings, zero-shot or few-shot classification, and optional IPC/CPC/WIPO-code validation, Bosphorus v0.1 offers a flexible starting point for more contextual patent intelligence systems.

Further development should focus on expert validation, threshold calibration, hybrid classification, and empirical benchmarking. Although experimental, Bosphorus v0.1 provides a practical foundation for moving beyond single-axis technical classification toward richer, application-aware patent analytics.

DECLARATION ON THE USE OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this working paper, generative AI tools were used to support English translation, language editing, stylistic refinement, and structural improvement. The author reviewed, edited, and approved the final text and takes full responsibility for the content, arguments, interpretations, and conclusions presented in the paper.

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